

OmniGroom Platform

Comprehensive Product Requirements & Technical Master Document

A unified, multi-tenant SaaS architecture defining the complete operational, financial, and user-experience frameworks for salons, barbershops, day spas, animal grooming facilities, and massage parlors.

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- Primary Stack:** Supabase, Render, Vercel, React Native
 - Communications:** Twilio (SMS), Firebase/Expo (Push)
 - Subscription Model:** Flat R65.00 ZAR / Month / Facility
 - Payment Gateways:** Payfast (Online) & 1Voucher (Prepaid)
 - Client Payments:** Strictly Offline / In-Store

1. EXECUTIVE SUMMARY & CORE OBJECTIVES

The platform serves as a centralized operating system for grooming and wellness businesses. It connects facility owners, their staff, and their clients through a cohesive digital ecosystem. The architecture is explicitly designed for multi-tenancy, allowing ambitious owners with multiple facilities to manage all locations from a single, centralized control panel.

Fundamental Workflow: Facilities manage their catalog, staff, and schedules via a mobile application. Clients discover services, track appointments, and communicate via a corresponding client mobile app. All online monetary transactions are restricted strictly to B2B subscription payments (R65/month), ensuring the platform remains free of complex C2B payment escrow liabilities.

2. COMPREHENSIVE FUNCTIONAL REQUIREMENTS

2.1 User Authentication & Onboarding

- **Omnichannel Sign-In:** Both the Client App and the Business App must support authentication via **Mobile Numbers (OTP via Twilio)**, **Google Accounts (Gmail)**, and **Facebook Login**.
- **Role-Based Access:** A single user account can hold different roles (Client, Staff, Owner).

2.2 Multi-Facility Centralized Management

- **The Master Dashboard:** An owner with multiple locations (e.g., a salon branch and a separate day spa branch) logs into one central account.
- **Seamless Context Switching:** The owner can toggle between facilities via a dropdown, instantly reloading the specific location's appointments, staff availability, and financials without logging out.
- **Isolated Billing:** Each facility operates as an independent tenant with its own R65/month subscription status. If one facility's subscription lapses, the others remain fully operational and visible to clients.

2.3 Service Catalog & Pricing Engine

- **Predefined Master Catalog:** Facility owners select the base services they offer from a global, predefined list (e.g., "Deep Tissue Massage", "Dog Wash").
- **Custom Overrides:** Once selected, the owner **MUST** set their specific price, duration, and optionally upload a custom representative image.
- **Image Fallbacks:** If the owner does not upload an image, the system automatically provides a high-quality temporary/default image from the predefined catalog.

2.4 Staff & Schedule Orchestration

- **Working Hours:** Facilities establish global operating hours. Individual staff members can have specific shift hours that fit within or override the global hours.
- **Service Assignment:** Owners assign specific services to specific staff members based on their skills.
- **Dynamic Availability:** The booking engine mathematically calculates available slots by cross-referencing facility hours, specific staff hours, and existing appointments to prevent double-booking.

2.5 Promotions & Marketing Media

- **Promotional Engine:** Facilities can run time-bound promotions, generating codes that clients can apply when booking.
- **Capped Gallery Uploads:** Facilities can upload portfolio images for potential clients to view. However, uploads are strictly restricted to a defined quota to manage storage costs. The backend must enforce this limit before accepting the file.

2.6 Communication & Notifications

- **Client Tracking:** Potential clients can track their upcoming appointments natively within the app.
- **SMS Verification & Alerts:** Twilio handles critical path SMS, including login OTPs and high-priority appointment reminders.
- **Push Notifications:** Firebase Cloud Messaging (FCM) or the free Expo Push Engine handles rich, real-time alerts for booking confirmations, cancellations, and inbound messages.

2.7 Omnichannel Meta Inbox

- **Centralized Messaging:** Facilities must be able to receive customer inquiries originating from Instagram and Facebook directly into the business app.
- **Webhook Routing:** The Render backend listens for Meta Graph API payloads, maps them to the correct facility, and triggers a push notification to the owner/staff, who can reply natively from the app.

3. FINANCIAL & SUBSCRIPTION ARCHITECTURE

3.1 Client-to-Facility Payments (Offline)

The platform acts purely as a scheduling engine. Clients book appointments without an online payment gateway. When the service is completed, the facility marks the appointment as "Completed" in the app and logs the offline payment method utilized on-site (Cash, Card/Speedpoint, or EFT).

3.2 Facility-to-Platform Subscriptions (R65 Flat Rate)

Platform access is gated by a strict, single-tier pricing model of **R65.00 ZAR per month, per facility**. This is managed by platform admins globally.

- **Digital/Banked Route (Payfast):** Owners can link a card or use instant EFT via Payfast for automatic monthly recurring billing.
- **Prepaid Cash Route (1Voucher):** Unbanked owners or those preferring cash can purchase a 1Voucher at retail stores. They input the 16-digit PIN into the app. The backend validates the PIN via the Flash API. If the value equals R65, the facility's subscription is extended by 30 days.
- **Enforcement:** Facilities with expired subscriptions are automatically hidden from the client-facing discovery app until renewed.

4. TECHNICAL STACK & INFRASTRUCTURE

Layer	Technology	Responsibility
Database & Auth	Supabase (PostgreSQL)	Data persistence, Multi-tenant Row Level Security (RLS), Google/Facebook/OTP Auth integration.
Backend API Core	Render (Node.js)	Webhook receivers for Meta/Payfast, 1Voucher API validation, background cron jobs for subscription states.
Web & Landing Page	Vercel (Next.js)	Public SEO marketing site, feature breakdown, download links, and initial account registration portal.
Mobile Applications	React Native (Expo)	Shared cross-platform codebase for both the Client App and the Business Management App (iOS & Android).
Communications	Twilio & Firebase	Twilio for transactional SMS. Firebase (or Expo native) for push notifications.

5. UI/UX DESIGN LANGUAGE & AESTHETICS

The platform interfaces—particularly the Business App—must balance high data density (calendars, rosters, lists) with a premium visual experience.

- **Aesthetic Tone:** The layout style must be modern, elegant, and highly professional.

- **Color Palette:** The core branding relies on a sophisticated mixture of sweet pinks, deep purples (#ff00db), and dark blues.
- **Clarity Constraints:** To ensure absolute legibility of schedules and appointments, backgrounds must utilize a single, uniform color. Gradients or complex background patterns must be completely removed from functional workspace areas to maintain a sleek, unified look.
- **Visual Precision:** Elements should appear raised with light shadow formatting to create depth without clutter. Only explicit, high-quality images provided by the facilities should be used; generic AI-generated filler art is strictly prohibited.

6. CORE DATABASE SCHEMA (MULTI-TENANT)

Below is the architectural blueprint for the PostgreSQL schema, demonstrating the relationships that enable central management and isolated billing.

```

-- Core User Identity
CREATE TABLE profiles (
    id UUID REFERENCES auth.users PRIMARY KEY,
    phone_number VARCHAR(20) UNIQUE NOT NULL,
    full_name VARCHAR(100) NOT NULL,
    role VARCHAR(20) NOT NULL CHECK (role IN ('client', 'staff', 'owner', 'admin'))
);

-- Master Facility (Tenant)
CREATE TABLE facilities (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    owner_id UUID REFERENCES profiles(id) NOT NULL,
    name VARCHAR(150) NOT NULL,
    facility_type VARCHAR(50) NOT NULL,
    working_hours JSONB NOT NULL
);

-- Subscription Engine
CREATE TABLE facility_subscriptions (
    facility_id UUID REFERENCES facilities(id) PRIMARY KEY,
    status VARCHAR(20) DEFAULT 'trial',
    payment_method VARCHAR(20) CHECK (payment_method IN ('payfast', '1voucher')),
    current_period_end TIMESTAMPTZ NOT NULL
);

-- Service Overrides
CREATE TABLE facility_services (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    facility_id UUID REFERENCES facilities(id) NOT NULL,
    catalog_id UUID NOT NULL, -- References global predefined list
    price_zar NUMERIC(10, 2) NOT NULL,
    duration_minutes INT NOT NULL,
    custom_image_url VARCHAR(255) -- Null falls back to temporary image
);

-- Appointment Tracking
CREATE TABLE appointments (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    facility_id UUID REFERENCES facilities(id) NOT NULL,
    client_id UUID REFERENCES profiles(id) NOT NULL,
    staff_id UUID NOT NULL,
    service_id UUID REFERENCES facility_services(id) NOT NULL,
    start_time TIMESTAMPTZ NOT NULL,
    end_time TIMESTAMPTZ NOT NULL,
    status VARCHAR(20) DEFAULT 'pending',
    offline_payment_method VARCHAR(20)
);

```